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BRIEF REPORT

Rootless hair as a reliable source of forensic genetic information

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One Sentence Summary:Personal identification from rootless hair shafts.

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Abstract

The small amount of fragmented DNA present in rootless hair shafts is unsuitable for many assays, including most PCR-based assays. Thus, rootless hair DNA is often overlooked in biomedical and forensic analysis. Here we apply methods for efficient recovery and sequencing of minute quantities of short, fragmented DNA to single human hair shafts. Using this approach, we characterize DNA fragments in hair shafts. Despite the small quantities of fragmented DNA, we find multi-fold genome coverage can be generated from a few centimeters of most hair shafts – enough to generate accurate genotype calls and statistically compelling evidence for identity or non-identity.

Background

DNA-based identification from forensic samples is common practice in criminal justice. The most widely adopted approach is PCR amplification of a few dozen specific multi-allelic short tandem repeat (STR) sites. When properly generated, these data can be used to search criminal offender databases to identify matching genotypes (if present in the database) or, occasionally, genotypes similar enough to be close relatives [1]. This approach is so ubiquitous in the field of forensics that forensic DNA-profiling is generally synonymous with a genotype at these STR sites. This approach has several advantages. First, access to STR profiles in criminal offender databases, e.g., CODIS in the U.S., is carefully maintained [2]. Second, the STR sites were chosen to minimize [3] – but do not eliminate [4] - information that could be exploited for purposes other than personal identification. Finally, a fast, sensitive and extensively validated assay is available for genotyping STRs [5]. However, many crime-scene STR-based profiles do not identify a suspect because the individual is not in the criminal offender database [6].

Recently, a new approach for DNA-based forensic investigation has emerged: investigative genetic genealogy (IGG) [6–8]. DNA from forensic samples can be used to generate single nucleotide polymorphism (SNP) profiles at hundreds of thousands of positions using standard genotyping arrays or direct whole-genome sequencing [8, 9], provided there is DNA of sufficient quality and quantity. The genotype information can then be used to search existing databases of user-contributed genetic profiles. Unlike an STR-based search of a criminal offender database, the purpose of this search is to find genetic *relatives* of the focal individual. Through genetic genealogy analysis, this list of relatives can provide investigative leads for cases that may otherwise be intractable. This approach has provided leads to help solve many cold cases where other approaches did not [10, 11]. Further, the same DNA sequence data produced from an evidentiary sample can be used for comparison to a suspect sample, should one be developed. Several SNP-based approaches have been developed for testing identity [12–15].

SNP-based analysis presents new challenges and offers new opportunities for forensic DNA. One example of both is that samples with highly fragmented DNA that are not amenable for PCR-based STR analysis may nevertheless yield useful SNP information. Rootless hair shafts have typically been used only for mitochondrial analysis due to their poor performance in PCR-based nuclear STR analysis. Despite the poor STR performance, it is widely known that fragmented nuclear DNA is present in hair shafts [16] – even those that are hundreds [17–19] or thousands of years old [20, 21]. However, hair shafts have not been comprehensively characterized as a source for SNP-based forensic identification.

Here, we describe a simple, efficient, and automatable workflow for conversion of the fragmented DNA recovered from single hair shafts into libraries for DNA sequencing. While the recoverable amount of DNA varies amongst individuals and amongst hairs from the same individual, we find that both head and pubic hair shafts are reliable sources of information on nuclear SNPs and mitochondrial DNA. We surveyed the amount and characteristics of the DNA fragments recoverable from hair shafts. We find that any single hair is likely to harbor sufficient DNA to call genotypes and to identify or rule-out identity when compared to a SNP genotype generated using independent data. This approach provides a technical path for using the DNA present in rootless hair shafts for forensics and biomedical analysis.

Results and discussion

We collected head hair (N=50), pubic hair (N=30), and saliva (N=50) from 50 anonymous human volunteers. To process the saliva, we extracted DNA and generated Illumina sequencing libraries following standard approaches (Supplemental Methods). To pre-process the hairs, we removed the roots, if present, trimmed head hairs to 5 cm and pubic hairs to 3 cm and calculated each hair's volume.

For each pre-processed hair, we performed DNA extractions and generated Illumina sequencing libraries using methods optimized for degraded DNA (see "Methods"). We digested each hair separately using a lysis buffer developed for hair [22] and isolated the DNA following a silica-coated magnetic bead approach optimized for degraded DNA [23]. When developing the workflow, we found most single-hair DNA extractions yield DNA quantities below the level of fluorimetry detection. Therefore, we proceeded to prepare sequencing libraries without sacrificing any DNA extract for quantification. From each single hair extract, we generated three Illumina sequencing libraries using a single-stranded library protocol [16], which has been further optimized for small quantities of fragmented DNA [24]. The library protocol involves no further DNA fragmentation or end-polishing, retaining the native ends of each DNA fragment.

We sequenced roughly 100 million reads from each hair-derived DNA library (Additional file 2: Table S1). A portion of the saliva-derived DNA was converted into sequencing libraries and sequenced to ten-fold average genome coverage (Aditional file 1: Fig. S1). Separately, another portion of the saliva DNA was used for genotyping using the Illumina Infinium Global Screening Array v3.0.

DNA fragments in rootless hair shafts are the degradation products of the endonuclease DNase1L2 [25, 26]. After fragmentation, these DNA fragments apparently remain in place within elongating hair shafts and can be visualized using standard DNA stains [27]. In this way, and based on characteristics described below, the DNA present in hair shafts can be described as cell-free DNA. Our library preparation method, which does not alter the five-prime or three-prime ends, allows us to characterize the fragmentation patterns of the hair-derived DNA strands (Fig. 1A-D).

We found that both nuclear and mitochondrial DNA fragments found in hair shafts are short. In all samples, nearly all DNA fragments were less than 100 nucleotides. The high degree of fragmentation of hair shaft DNA, even within freshly cut hair shafts, is likely the culprit for the general failure of standard PCR amplification of STRs when attempted with DNA extracted from rootless hair. Nuclear DNA is present in hair shafts. However, the small quantity of recovered DNA is generally too short for PCR-based STR amplification where amplicons are typically several hundreds of base-pairs long (Additional file 1: Fig. S2).

DNA fragments in hair shafts showed other interesting and practically useful characteristics. Like the cell-free DNA fragments found in blood plasma [28], hair DNA fragments are enriched for genomic regions likely to be associated with histones (Additional file 1: Fig. S3). DNA fragments from the mitochondrial genome, which are not organized on histones, do not show a multi-modal fragment length distribution (Fig 1C and D). Unlike blood plasma cell-free DNA fragments, which are mostly complete mono-nucleosome DNA segments, hair DNA fragments are too short to comprise complete mono-nucleosomal DNA as was shown previously for DNA derived from very old hair [29]. Genetic sex determination is trivially done by comparing the ratio of DNA fragments that map to chromosome X and those that map to chromosome 10, a similarly sized autosome (Additional file 1: Fig. S4).

To further explore the fragmentation characteristics of DNA found in hair shafts relative to inferred nucleosome positions, we used the panel of all mapped hair DNA to infer the genome locations of 586,808 nucleosomes on chr12 based on coverage peaks in these data [30]. Using this nucleosome map, we compared the 5-prime and 3-prime end positions of reads mapped to chr12 relative to the closest nucleosome (Additional file 1: Fig. S5). This analysis revealed the 5-prime DNA end points most often occur 73 basepairs from the inferred center-point of each nucleosome. This observation is consistent with hair DNA fragmentation occurring on histones as histones are known to physically interact with 147 basepairs of DNA [31]. 3-prime DNA end points occur closer to the center-point of nucleosomes. A subtle but important feature is revealed by this analysis. Mapped DNA fragments show a clear and symmetrical strand bias around nucleosomes. DNA recovered from hair shafts tends to be from the strand that enters the nucleosome in the 5-prime to 3-prime direction. Note that these are opposite strands for each end of the histone.

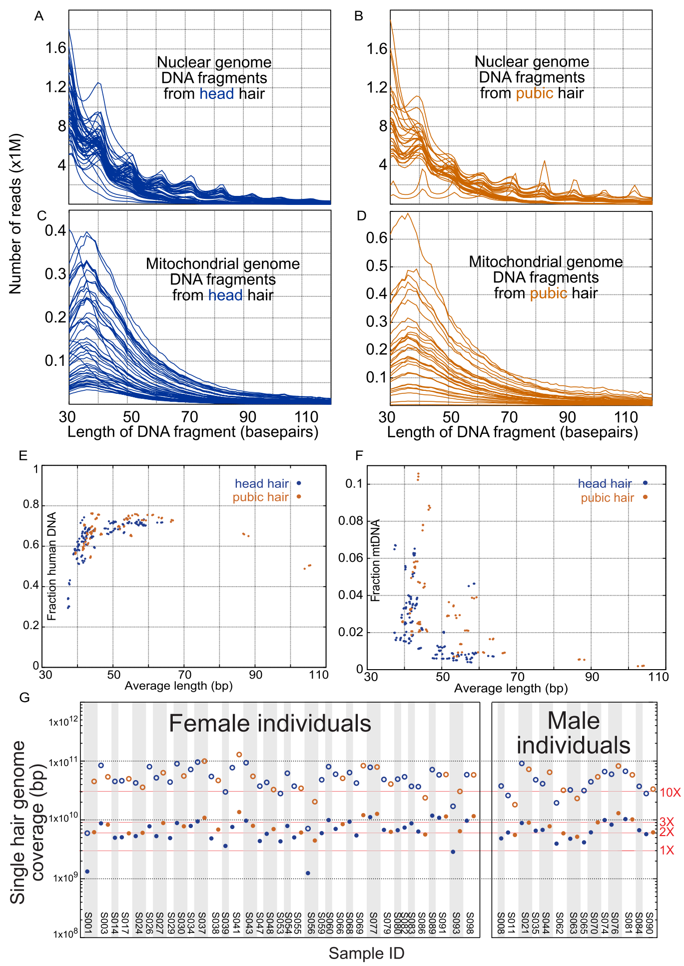

Fig. 1 Characteristics of DNA fragments recovered from single hair shafts – **A** and **B** the length distribution of DNA fragments that map to the human nuclear genome from each head hair (blue) and pubic hair (brown). **C** and **D** Length distribution of DNA fragments that map to the mitochondrial genome. **E** – **G** Each point represents the DNA sequence from a single library. Blue points are data from head hairs. Brown points are data from pubic hairs. **E** The fraction of DNA reads from each hair that map to the human genome. **F** The fraction of human mapping reads that map to the mitochondrial genome. Note, most human-mapping DNA maps to the nuclear genome. **G** The achieved coverage, in base pairs, for each hair (closed circles) and estimated achievable coverage (open circles) if all unique molecules in the libraries were sequenced

Hair DNA fragments also show a strong sequence bias around the fragmentation sites at both the five-prime and three-prime ends (Additional file 1: Fig. S6). This fragmentation bias could reflect cleavage biases in DNase1L2 [25] or other active nucleases [32], a sequence bias at preferred fragmentation sites within nucleosomes, sequence motifs enriched at the end-points of histone-bound DNA [33], or a combination of these influences.

We generated similar amounts of raw data from the sequencing libraries of each hair (Additional file 2: Table S1). Nevertheless, the amount of usable DNA information differed due to (1) the fraction of human DNA present in the library (Fig. 1E), (2) the ratio of nuclear to mitochondrial DNA within the human DNA component (Fig. 1F), and (3) the length distribution of human DNA fragments (Fig. 1A-D). In all cases, the majority of human DNA recovered from hair shafts was from the nuclear genome (mean 96.94%) with a small, but variable minority of the human DNA from the mitochondrial genome (mean 3.06%) consistent with another recent analysis of DNA from hair [16]. The variability in human DNA recovered was not well predicted by the volume of each hair (Additional file 1: Fig. S7)

Overall, we were able to achieve an average of 2.18 fold genome coverage per head hair (range 0.40–3.86) and an average of 2.64 fold genome coverage per pubic hair (range 1.45–4.47) from about 300 million total reads from each single hair (Fig. 1G). Despite the variable fraction of mitochondrial DNA across hairs, every hair library produced a mitochondrial consensus genome with the same haplotype as that produced from the higher-coverage saliva DNA (Additional file 1: Fig. S8). The diversity of the mitochondrial tree and PCA analysis of the panel individuals projected onto 1000 Genomes populations (Additional file 1: Fig. S9), reveals the genetic diversity present amongst the 50 anonymous study members in the panel.

Next, we assessed the utility of DNA recovered from single hairs to provide statistical evidence for or against identity. We compared the DNA sequence from each single hair to genotypes called from each of the individual saliva samples. First, we performed variant calling on the saliva DNA using GATK [34], GLIMPSE2 [35], and a custom-built genotype caller designed for low coverage data, as described in the SOM. We generated calls on all known variable sites from the 1000 Genomes database [36]. The final call sets for GATK and GLIMPSE2 contain 83,787,041 and 59,958,132 total variants, respectively. We then filtered the genotype calls generated by GATK and GLIMPSE to a set of about one million bi-allelic sites found in the union of several common direct-to-consumer platforms whose genomic coordinates could be unambiguously determined and have a minor-allele count of at least 3 in the 1000 Genomes database. We then applied a GQ threshold of 20 for each genotype call.

We used IBDGem [24] to calculate a likelihood ratio that can inform whether the hair DNA comes from a specific individual or not. IBDGem calculates the likelihood of an input DNA dataset, from a hair, for example, under the model that it derives from a specific comparison genotype. Separately, it calculates the likelihood of the input DNA data under a model that the DNA derives from an unknown, unrelated individual as modeled by genomic segments of the genotypes in the 1000 Genomes Project individuals [37]. The ratio of these likelihoods is then reported by IBDGem.

We ran this program for all pairwise comparisons between the 80 hairs and the 50 saliva-derived genotype files. This comprises 80 positive-control tests and 3,920 negative control tests (Fig. 2). The log-ratio statistic reported is for the chromosome arm that least strongly supports identity between the hair DNA and the saliva-derived genotype. Regardless of the program used to call genotypes, all positive control and negative control comparisons provided strong evidence for the correct answer: identity or non-identity. This was true when all data generated per hair was used (Fig. 2A, B) or when hair DNA data was down-sampled to less than 0.2 fold coverage (Fig. 2C, D). Similar results were generated regardless of the GQ threshold used for the comparison genotypes or the genotype caller (Additional file 1: Figs. S10-S12).

We note that in all cases IBDGem produced a statistic that strongly supported the correct hypothesis, regardless of whether the correct hypothesis was identity or

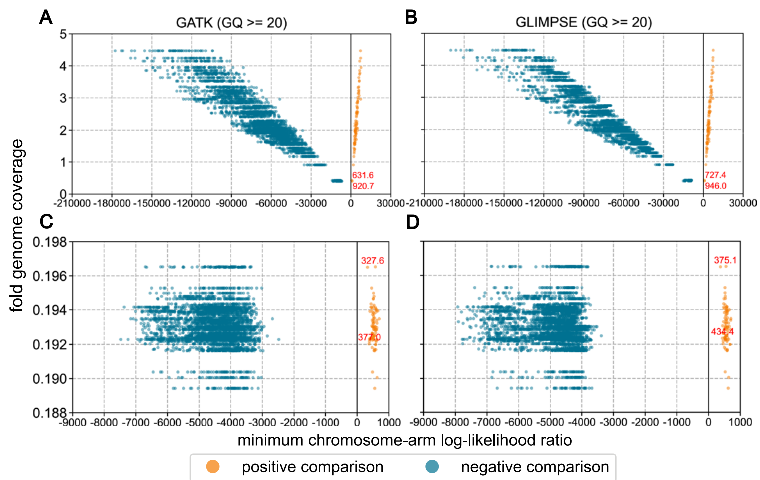

Fig. 2 IBDGem comparison of hair data to known genotypes. Genotypes for each of the 50 individuals were generated using DNA from saliva using GATK (left) or GLIMPSE (right). Data from each hair was compared to the saliva-derived genotype from the same individual (orange/positive comparisons) or to the genotype of each of the 49 other individuals. X-axis values are the log base2 exponent of the ratio of the likelihood of the data under the model that they derive from the comparison genotype to the likelihood of the data under than model that they derive from an unknown, unrelated individual. The reported value is the aggregated log likelihood ratio of the chromosome arm that least confidently supports the model that the data derive from the comparison genotype. In each panel, the x-axis values for the two positive control tests with the lowest chromosome-arm log likelihood ratios are labeled in red. **A** Comparisons with GATK called genotypes, filtered at GQ>=20. **B** Comparisons with GLIMPSE called genotypes, filtered at GQ=>20. **C** Randomly down-sampled hair data with GATK genotypes as in (**A**). **D** Randomly down-sampled hair data with GLIMPSE genotypes as in (**B**)

non-identity. Of note is that in the non-identity/negative control tests, the strength of the statistic was uniformly more extreme than in the identity/positive control tests. That is, negative control tests are more negative than positive control tests are positive. Because the test statistic is the chromosome arm that least confidently supports identity (or equivalently, most confidently supports non-identity), this could be partially due to the test identifying different chromosomes to report in the positive versus negative control experiments. To assess whether this conservative asymmetry is a general feature of IBDGem identity comparisons with these data, we analyzed the log likelihood ratios of the negative control tests using whatever chromosome arm was reported in the positive control tests for each comparison genotype (Additional file 1: Figs. S13 and S14). In these tests, negative controls are also consistently more negative than positive control tests.

In an investigative setting, DNA from an evidence sample can also be used to generate genotype files for investigative genetic genealogy (IGG). We used the data from each hair to generate genotype calls at sites in the most widely used direct-to-consumer genotype arrays. As expected, the number of sites called and the concordance between lower coverage hair data versus higher coverage saliva data both increased as a function of the amount of DNA sequence from the hair (Additional file 1: Fig S15). For the 77 hairs that generated greater than 1-fold average coverage, genotype concordance was > 99.4%. In practice, this level of accuracy is sufficient to identify genetic relatives.

Conclusions

The ubiquity of shed hair means that many crime scenes contain hairs when other forensic evidence may be absent. Further, past efforts to compare hair morphology and other microscopic features motivated the collection of hair from crimes that remain unsolved. In practice, we have used the approach described here to help generate investigative leads for criminal investigations and identification of human remains. Thus far, this approach has only been used by us for law enforcement purposes in violent crime cases and in analyzing DNA of deceased people. While using this approach for other purposes would violate the Terms of Service of the databases required to identify genetic relatives, stronger disincentives should be considered. When combined with direct comparisons that can provide strong statistical support for identity, it is now possible to find suspects and test identity from the DNA present in a few centimeters of a single rootless hair.

Methods

Sample Processing

For each collected hair, we first examined and removed identifiable roots. If the hair lacked an identifiable root, we trimmed approximately 1 cm from each end. Following root removal, we trimmed the head hairs to 5 cm and pubic hairs to 3 cm or recorded the length of hairs shorter than the desired length. We imaged each hair using a Zeiss Primo Star microscope and SwiftCam microscope camera. Following calibration using the provided reference slide, we recorded the diameter of each hair using the SwiftImaging 3.0 software. Finally, we calculated the approximate volume of each hair using the measured length and diameter.

Hair Processing – DNA Extraction

From the 50 individuals, we extracted and isolated DNA from up to 2 pre-processed rootless head hairs (up to 5 cm) and 1 rootless pubic hair (up to 3 cm), for a total of 129 DNA extractions. To clean the hair exterior, we submerged each hair in 0.5% sodium hypochlorite for 10 seconds and then submerged the hairs in three water baths for 10 seconds each. We placed each cleaned hair in 145 μL lysis buffer (2% SDS, 10 mM Tris-HCl, 2.5 mM EDTA, 10 mM NaCl, 5 mM CaCl2, 40 mM DTT, and 2 mg/mL Proteinase K) and incubated the hairs overnight in a 750 rpm, 55 °C thermomixer. Following the overnight incubation, we added 3 μL of 1M DTT and 7 μL of 20 mg/mL Proteinase K and incubated each sample for an additional 1 h in a 750rpm, 55 °C ThermoMixer. Next, we cooled each sample on ice for 5 m and spun the tubes for 2 m at 16,000 rcf in a microcentrifuge. Finally, we transferred 150 μL of the supernatant to a new tube while avoiding the pellet.

We isolated DNA from 150 μL of the supernatant following the silica magnetic bead approach as described in Rohland et al. [23] using Buffer D and eluted in 28 μL of buffer EBT (10 mM Tris, 0.05% Tween-20).

Hair Processing – Library Preparation

We exhausted each DNA extraction by preparing three Illumina libraries per hair DNA extract, each using one-third of the extract. This generated a total of 387 libraries. We prepared single-stranded DNA Illumina libraries as described in Kapp et al. [38] with the reagent concentration modifications described in the supplemental information of Nguyen et al. [24]. Briefly, we combined 9 µL of DNA extract with 2 µL 38 ng/µL ET SSB (NEB) and heat denatured the DNA for 3 m at 95 °C in a Thermocycler. Next, we cold-shocked the libraries for 2 minutes and added 2 µL of an adapter mix containing 1 µM P5 splinted adapter and 0.2 µM P7 splinted adapter. Following adapter addition, we added 12 µL of a reaction mix containing 9.20 µL 50% PEG 8000, 1.25 µL 1M Tris-HCl, 0.25 µL 1M MgCl2, 0.25 1M DTT, 0.25 µL 100 mM ATP, 0.55 µL 10,000 U/µL PNK (NEB), and 0.25 µL 2,000,000 U/µL T4 DNA Ligase (NEB) and incubated the libraries at 37 °C for 1h. Finally, we performed a SPRI clean as described in Rohland and Reich [39] using 35 µL buffer EBT and 60 µL SPRI solution for the initial incubation and 21 µL buffer EBT for elution. We developed automation protocols and performed the pre-PCR liquid handling steps on an Agilent Bravo with an NGS option A configuration.

We performed qPCR on each unamplified library using primers IS7 and IS8 described in Gansauge and Meyer [40] to inform the optimal cycle number for each library during indexing PCR. We prepared 50 µL qPCR reactions containing: 1 µL library, 25 µL 2X Maxima SYBR Green Master Mix (Thermo Scientific), 0.5 µL 100 µM primer IS7, 0.5 µL 100 µM primer, and 23 µL H2O. We cycled the reactions using a BioRad CFX Opus 96 with the following conditions: 95 °C for 10 m, followed by 40 cycles of 95 °C for 30 s, 60 °C for 30 s, and 72 °C for 30s. Fluorescence was measured at the end of each extension step.

Following qPCR, we amplified and double-indexed each library using 384 unique i5 and 384 unique i7 indexes following the oligo design described Kircher et al. [41]. We prepared 50 µL indexing reactions containing: 20 µL library, 25 µL 2X Amplitaq Gold 360 Master Mix (Applied Biosystems), 2.5 µL 20 µM i7 indexing primer, and 2.5 µL 20 µM i5 indexing primer. Next, we amplified the libraries with the following conditions: 95 °C for 10 m, followed by a library-specific number of cycles of 95 °C for 30 s, 60 °C for 30 s, and 72 °C for 60 s, and a final extension of 72 °C for 7m. We purified the amplified libraries with a SPRI ratio of 1.2X and eluted the cleaned libraries in 22 µL buffer EBT.

Finally, we quantified the amplified libraries using a Qubit 1X dsDNA HS Assay Kit and a Qubit Flex fluorometer before pooling equal-nanogram for sequencing.

Saliva Processing

From the 50 individuals, we extracted and isolated DNA from saliva using the prepIT-L2P reagent and protocol (DNA Genotek). We incubated the saliva collection tubes for 2 hours in a 50 °C, 500 rpm Thermomixer. Following incubation, we aliquoted 500 µL of material into 1.5 mL tubes and followed the isolation protocol outlined in the prepIT-L2P manual. Finally, we eluted the DNA in 100 µL of buffer TE (10 mM Tris-HCl, 1 mM EDTA) and then incubated the extracts for 1 hour in a 50 °C, 500 rpm Thermomixer to rehydrate the DNA.

From 44 of the 50 individual saliva extracts, we submitted approximately 750 ng of DNA for genotyping to the University of California, Berkeley Genetic Epidemiology and Genomics Lab using the Illumina Infinium Global Screening Array. We chose 44 because there is a 48-sample array format and 4 spots were reserved for controls.

From all 50 saliva extracts, we prepared Illumina libraries using the NEBNext Ultra II FS DNA Library Prep Kit (NEB). We followed the protocol as described with 200 ng of DNA input, 6 minutes of enzymatic shearing time, and amplified with dual unique indexes for 4 cycles. Finally, we purified the amplified libraries using a SPRI ratio of 1.2X and quantified using a Qubit 1X dsDNA HS Assay Kit and a Qubit Flex fluorometer before pooling equal-nanogram for sequencing.

Sequencing

For quality control, we sequenced all libraries at the University of California, Santa Cruz Ancient and Degraded DNA Processing Center on an Illumina NextSeq 550 using 150 cycle mid-output reagent kits.

For deeper sequencing, we submitted libraries to the Duke Center for Genomic and Computational Biology for sequencing on a NovaSeq 6000 using S4 200 cycle reagent kits. We submitted the saliva libraries for 1 flow cell, where each library was pooled to receive 1/50th of the reads. Additionally, we submitted the 3 libraries associated with 1 hair extraction and 1 pubic hair extraction for 3 additional S4 flow cells, where each library received about 300,000,000 reads.

Sequence alignment

Raw sequence read pairs from hair were merged using SeqPrep2. Because the DNA fragments from hair are generally shorter than the read length, nearly all read pairs were meged into a single read representing the entirety of each DNA fragment. Merged reads whose length is >= 30 nucleotides were kept for further processing. These merged reads were aligned to the human genome (GRCH38) using bwa aln 0.7.17-r1188 using option -l 24.

Sequence data from saliva-derived libraries was aligned using bwa mem version 0.7.17-r1188 with -L10,10 flag values.

Per-library PCR and optical duplicates were marked in each bam file using picard MarkDuplicates Version:2.25.7

To generate down-sampled sequence data, we first down-sampled the BAM file to the target coverage using samtools view, then converted the output to pileup format via samtools mpileup.

Genotype calling from saliva-derived sequence data

GLIMPSE2 - We generated genotype posteriors from the bam files generated from the spit data with glimpse2 - v2.0 [35], using the high coverage 1000 genomes NYGC re-release [37] as our reference panel. We performed all reference panel creation and subsequent imputation steps using default settings. Briefly, we called glimpse_chunk to identify optimal imputation chunks from the reference panel before calling glimpse_split_reference to construct the binary reference panel from the 1000 genomes VCF. We then called glimpse_phase to impute variants in our bam files and phase haplotypes before calling glimpse_ligate to construct VCF files for each chromosome from the imputed variants in each window. Since GLIMPSE generates genotype posterior probabilities (GP) but not genotype quality (GQ) scores, we calculated GQ by first converting GP to GL (genotype likelihoods). Then, we converted GL to PL (Phread-scale genotype likelihoods) using bcftools +tag2tag. Then, the corresponding GQ value is calculated by taking the different between the second lowest and lowest PL score for each call, as implemented in GATK's HaplotypeCaller.

GATK – We generated genotype calls using GATK version 4.6.1.0 [34] via the following set of commands: BaseRecalibrator (using 1000 Genomes sites), ApplyBQSR, IndexFeatureFile (using 1000 Genomes sites), HaplotypeCaller, CombineGVCFs (to make one master VCF file with all individuals), GenotypeGVCFs (joint genotype calling), VariantRecalibratior, ApplyVQSR.

Both GLIMPSE2 and GATK VCF files were then filtered for specific GQ thresholds and to the list of variant sites found on common direct-to-consumer platforms that could be confidently mapped to the GRCH38 version of the reference human genome as described in the "Marker selection for genotype files" section below.

IBDGem comparisons

We ran IBDGem v2.1.2-beta using genotypes from each of the 50 saliva-derived genotype files called by 3 different genotype callers. We compared the hair data from each of the 80 individual hairs. For each comparison, we used the hair DNA data (in Pileup format), along with two VCF files: one containing genotypes of all unrelated samples from the 1000 Genomes, and one containing the comparison individual's saliva-derived genotypes. These were used to calculate the likelihood under the model that the hair data originated from a random, unrelated individual versus the likelihood under the model that it came from the comparison sample. We applied the -T option to limit the comparison to the target sites found on common direct-to-consumer platforms (see "Marker selection for genotype files" section). We also used the -v option to exclude sites where the saliva genotype was homozygous reference. All other settings were left at their default values.

Following the comparisons, we computed the cumulative log base2 of the likelihood ratio (LLR) between the two models across each chromosome arm using the chrarm-stats helper script provided with IBDGem. Unless otherwise specified, the chromosome arm that least supports the hypothesis that the hair DNA and saliva genotype come from the same individual (i.e., the chromosome arm with the minimum LLR) was used as the reporting statistic for each comparison.

Genotype calling algorithm – astrea-impute2

To convert raw DNA sequence information into genotype files, we implemented a genotype caller that takes advantage of the unique aspects of the DNA fragments present in hair. In brief, this program examines the available hair DNA sequence data at each target site and nearby linked sites. The target sites were chosen to be the union of all bi-allelic single-nucleotide variant sites present on the most common direct-to-consumer genotype platforms since only these sites are useful for searching available genetic databases and identifying genetic relatives. The linked sites were chosen such that they contain maximal mutual information, i.e., linkage disequilibrium, for each target, are likely to be present in a DNA library from hair and are spaced such that no single DNA fragment will cover more than one of the linked sites. We precompute target haplotype information (target and linked sites) using all unrelated individuals in the 1000 Genomes panel into a simple, machine-parseable format. Genotypes are called by finding the likelihood of the observed DNA sequence data at each site in the haplotype summary. This approach is parallelizable across target sites and was thus implemented as a multi-threaded executable. It calls genotypes from aligned sequence data in less than one hour on a 64-threaded machine.

For each target site, we pre-compute a haplotype summary of linked allelic variation at and around each target site. For these haplotype summaries, we use all the phased genotype information for all unrelated individuals in the phased, high coverage 1000 Genomes VCF [37]. For each target site, the surrounding haplotype region considered is arbitrarily set at 5,000 base-pairs up and downstream of the target site. Within this region, we examine each 100 base-pair window for the single most useful linked variant site.

Useful for the purpose of calling genotypes from data present within rootless hair is defined by the following criteria:

(1) transversions are preferable over transitions
(2) linkage disequilibrium between the site under consideration and the target site
(3) between 50% and 200% of the average fold coverage at that site across all hair data in the panel.

100 base-pair windows are chosen to minimize the chance that any single read will contribute data to the overall genotype call.

Each target site haplotype summary, H, can be expressed as a set of genotype vectors, each vector representing the alleles present in an observed haplotype at the chosen sites around the target site. Each allele can be written as 0 (reference allele) or 1 (alternate allele). Thus, each haplotype vector represents the allelic variation present in one chromosome in the region of interest at the sites chosen. Because the total length of the region is 10 kb and at most one site is chosen per 100 base-pair window, the length of the vector is at most 100.

$$H = \left\{ \begin{array}{c} H_0 \\ H_1 \\ \ldots \end{array} \right\} = \left\{ \begin{array}{c} (0|1, 0|1, \ldots) \\ (0|1, 0|1, \ldots) \\ \ldots \end{array} \right\}$$

Note that for any pair of haplotype vectors, there is an implied genotype at each site in the haplotype summary, including the target site. That is, for any pair of haplotypes from H implies a specific unphased genotype.

$$\left(H_i, H_j\right) \Rightarrow G_{i,j}$$

The probability of observed data at site s (Ds) can be calculated given a genotype at site s. As described above, a genotype is implied at each site by any haplotype pair under consideration. If we represent the data at site s as the number of observed reference and observed alternate allele counts:

$$D_s = (D_{s,r}, D_{s,a})$$

Then the probability of the data, *Ds*, given the genotype *Gi,j,s* at site *s*, that is implied by the haplotypes *Hi* and *Hj* is:

$$Pr(D_s|G_{i,j,s}) = \begin{cases} \binom{D_{s,r}+D_{s,a}}{D_{s,r}}(1-\varepsilon)^{D_{s,r}}\varepsilon^{D_{s,a}};\ G_{i,j,s} = 0{,}0 \\ \binom{D_{s,r}+D_{s,a}}{D_{s,r}}\left(\frac{1}{2}\right)^{D_{s,r}}\left(\frac{1}{2}\right)^{D_{s,a}};\ G_{i,j,s} = 0{,}1 \\ \binom{D_{s,r}+D_{s,a}}{D_{s,r}}(1-\varepsilon)^{D_{s,a}}\varepsilon^{D_{s,r}};\ G_{i,j,s} = 1{,}1 \end{cases}$$

Because the target sites are selected such that independent data, i.e., reads, are present within each *Ds*, we calculate the probability of the data for each haplotype pair across all sites in the hapotype summary by the product rule. We use the observed haplotype frequencies as priors for each possible pair of haplotypes.

While there are $(n^2-n)/2$ unique haplotype pairs in each haplotype summary, each pair implies one of only 3 possible target site genotypes: 0,0 (homozygous reference), 0,1 (heterozygous) or 1,1 (homozygous alternate).

For each haplotype pair, we add the likelihood of the data to an aggregate likelihood of the implied target site genotype for that haplotype pair. Once all haplotype pairs are computed, we select the target site genotype as the one with the highest likelihood and report genotype probability scores and Bayes Factor scores for the genotype call at the target site.

Note that each target site haplotype summary is precomputed from the input reference panel. Thus, calling genotypes and their associated scores in parallelizable across target sites. The program is implemented as a multi-threaded executable, astrea-impute2.14. Average run time for data from each hair data on chr1 is 131 seconds on our two-socket Intel Xeon Silver 4216 (2.10 Ghz) server.

We called genotypes in VCF format using astrea-impute2.14 with -B 20 and -G 1 flags. The VCF output was then run as input to beagle5.2 [42] using the 1000 Genomes dataset as a reference panel. This allows for imputation of sites that were not called by astrea-impute2.14. We filtered the beagle5 output to remove genotype calls with GP < 0.99 and to remove calls that are discordant, i.e., no reads supporting imputed genotype call, with the sequence data present for that site.

Marker selection for genotype files

Sites of known genetic variation were chosen from the union of all sites on the Direct-to-Consumer array platforms AncestryDNA v2, 23andme v2 and v3, and the Illumina GSA v3 that have an rsID identifier and are not on chrMT or chrY (1,556,186 total sites). We then removed sites that had multiple, ambiguous mapping coordinates leaving 1,555,620 total sites.

We further filtered these sites using information from the UCSC browser snp151 tables mapped to both the hg38 and hg19 versions of the reference genome. Sites were removed if they failed any of the following filter criteria:

1. value in the "class" field is "single"
2. number of alleles is exactly two
3. both alleles are a valid, single base (A, C, G, or T)
4. site must be present in both hg38 and hg19 tables

Presence in both tables is necessary because the mapping and genotype calling framework is done using the hg38 reference genome. However, database searching in GEDMatch and other databases requires genotype files on the hg19 coordinate system. Therefore, genotype calls must be correctly transferred from hg38 to hg19 coordinate systems. We used the snp151 mapping information for each SNP for this transfer. After this filtering, 1,269,945 sites remained.

We then compared the strand field information for each site and made a list of sites whose genotype call requires complementing on hg19 coordinates versus hg38 coordinates.

Finally, we compared the minor allele frequency called from the 80 individual hair panel genotype files versus the minor allele frequency reported in the 1000 Genomes VCF file across all unrelated individuals. Sites that deviated by greater than 20% in the panel versus the 1000 Genomes value were further filtered from our call set as being unreliably callable given data recoverable from hair shafts using the methods described here and genotype arrays.

This left 1,013,317 total sites in our final callable set.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13059-026-03981-8.

Additional file 1: contains supplementary figures and Human Subjects protocol information.

Additional file 2: Excel spreadsheet of Table S1.

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Peer review information
Wenjing She was the primary editor of this article and managed its editorial process and peer review in collaboration with the rest of the editorial team. The peer-review history is available in the online version of this article.

Authors' contributions
Project was conceived by JDK, BS, and REG. JDK led laboratory procedures for data generation, assisted by HN, CW and TT. REG implemented software and led data analysis, assisted by RN, SS, CV, CS, KH. All authors assisted with writing and editing the manuscript. All authors read and approved the final manuscript.

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Data availability
Raw data for this project are archived in dbGap: phs002979 "Evaluation of Nuclear DNA from Rootless Hairs for Forensic Purposes" [43]. Data access can be requested by visiting https://dbgap.ncbi.nlm.nih.gov/home/. The genotype calling software, astrea-impute2.14 (v14) is available via Github: github.com/astreaforensics/astreaimpute2 and archived in zenodo (DOI:10.5281/zenodo.18462628) and is released under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Public License.

Declarations

Ethics approval and consent to participate
In compliance with the Helsinki Declaration, study data was collected under UCSC IRB approved Human Subjects Protocol HS3382 – Evaluation of nuclear DNA from rootless hair for forensic purposes.

Consent for publication
Not applicable

Competing interests
REG, JDK, and RN are inventors of technology described in this report. REG is an equity holder and paid consultant of Astrea Forensics. JDK is a paid consultant of Astrea Forensics. BS is currently CSO of Colossal Biosciences.

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